

Hidden keys and repeated biometric records

Can multiple protected records reveal identity when keys remain hidden?

Fresh-key learned attacks: uncertainty remains; no equivalence claim.

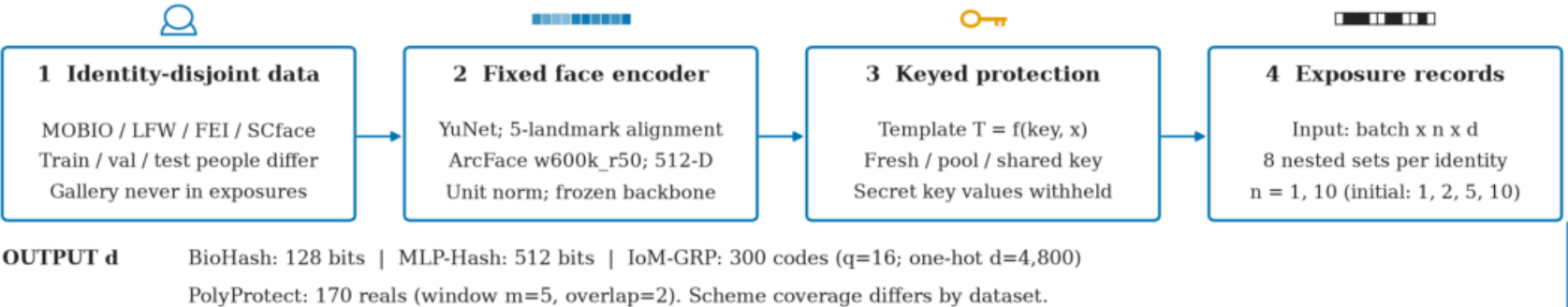
Recurring transforms: large gains from multiple same-identity records.

Boundary location changes with the dataset and identity partition.

Independent study. No exact benchmark reproduction or universal privacy claim.

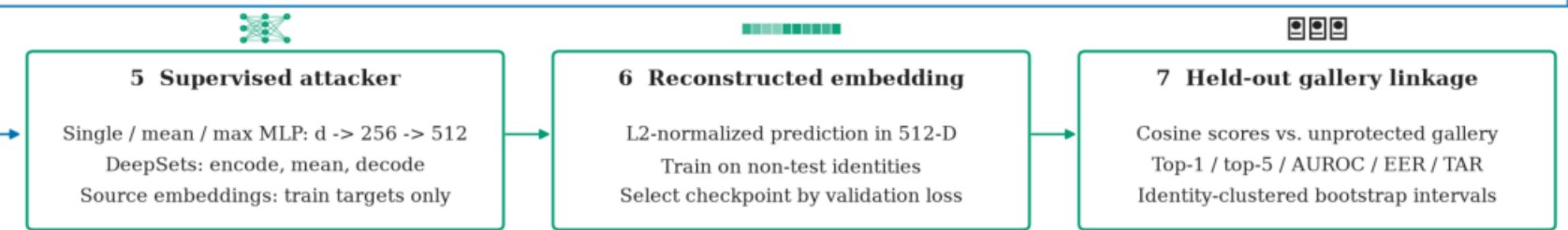
Experimental architecture

A DATA AND PROTECTION



B KEY-BLIND ATTACK AND EVALUATION

Attacker observes protected records, not keys or slot labels; slots are revealed only in a labelled control.



- KEY DESIGN** Fresh: source-record keys are split-disjoint. Pool K: hidden transforms recur across identity splits.
- UNCERTAINTY** Earlier studies: 3 model seeds. IoM-GRP / PolyProtect pilots: 1 seed; not confirmation.
- SCOPE** Tests transform reuse and same-identity aggregation, not a new backbone or a universal privacy guarantee.

Train on separate identities. Reconstruct an embedding, then link to a held-out gallery.

Dataset and protection coverage

Dataset	Identities	Train/val/test	Embeddings	Variation
MOBIO	150	90/30/30	1,799 / 1,800	Sessions
LFW	125	75/25/25	1,500 / 1,500	In-the-wild
FEI	200	120/40/40	2,378 / 2,400	Pose/expression
SCface	130	78/26/26	2,851 / 2,860	Camera/distance

BioHash: 128 bits; four datasets; includes a Haar-sign control on MOBIO.

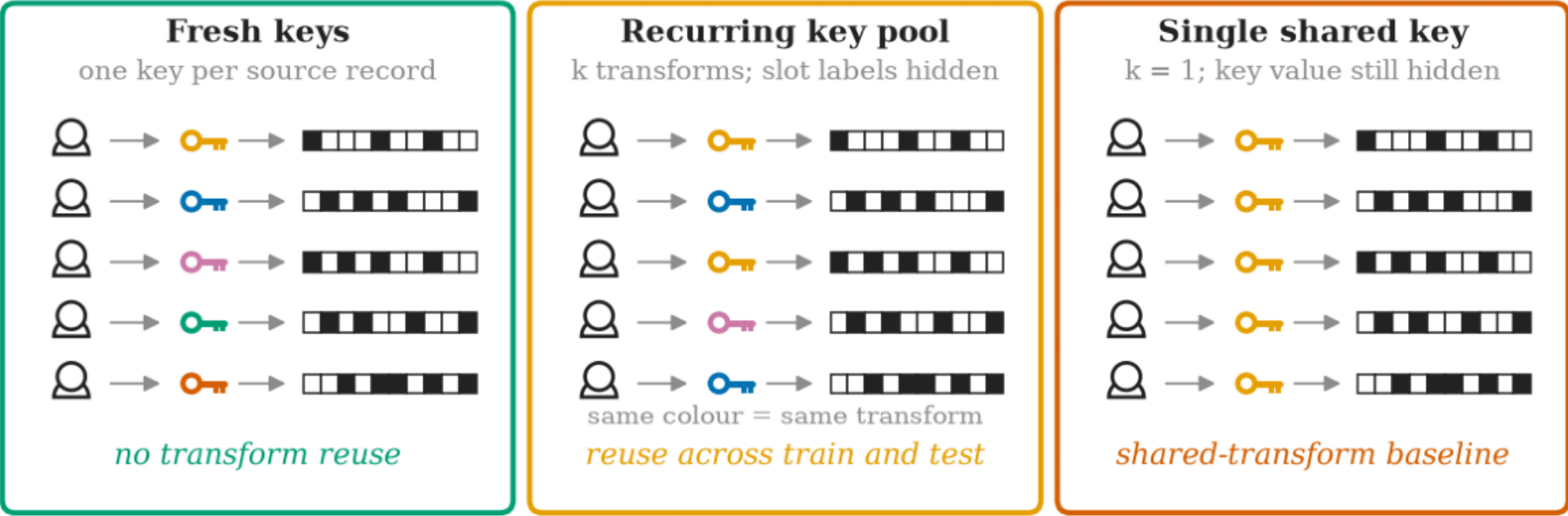
MLP-Hash: 512 bits; MOBIO; paper-specified, not source-exact.

IoM-GRP: 300 categorical codes (q=16); MOBIO/FEI/SCface pilots.

PolyProtect: 170 real values (m=5, overlap=2); MOBIO/FEI/SCface pilots.

SCface: mugshot gallery and visible surveillance probes; 9 detection failures, all identities eligible.

What changes between key regimes?

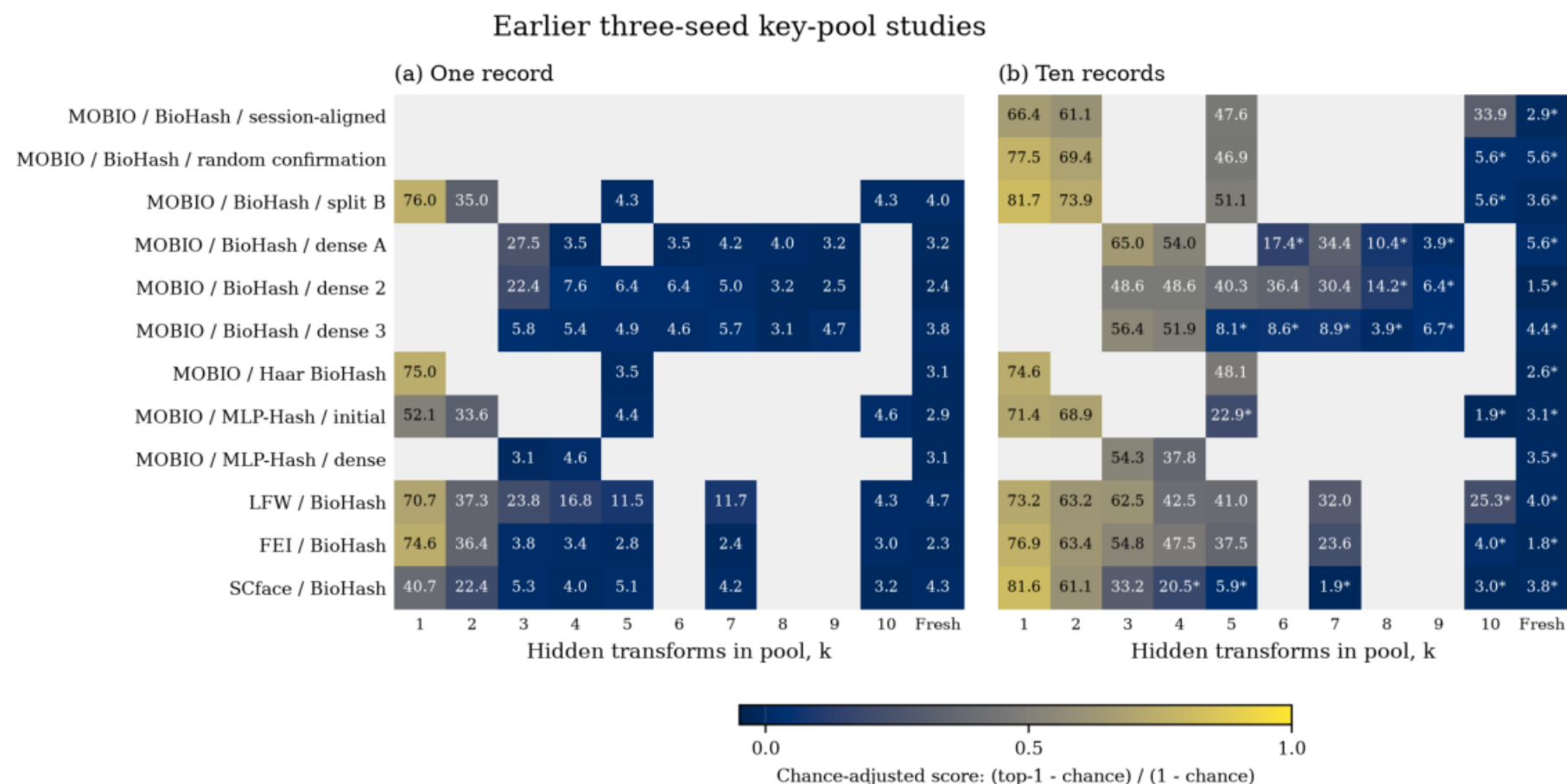


Learned attacks inconclusive **Multi-record gains observed** **Strong linkage observed**
PolyProtect outside the theorem boundary depends on dataset and split hidden does not mean unlearnable

Protocol Disjoint train / validation / test identities; one held-out gallery image per identity
8 sets per identity; 3 seeds earlier, 1 in pilots; identity-clustered intervals

Key values stay hidden in every regime. Recurring transforms are shared across identity splits.

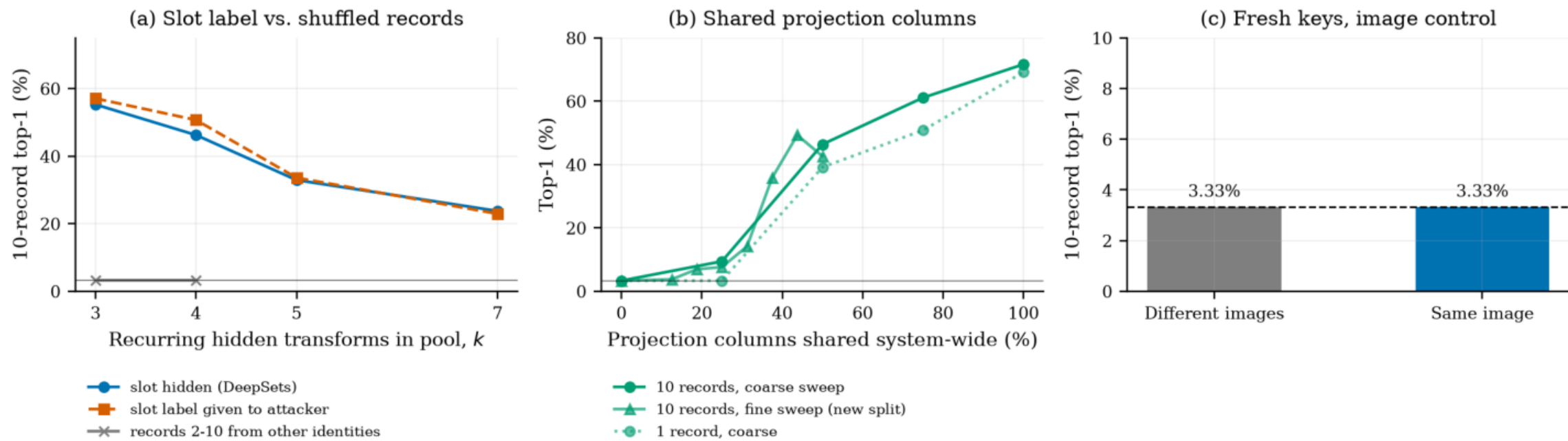
Earlier three-seed key-pool studies



73 conditions from 12 earlier studies. New one-seed pilots are reported separately.

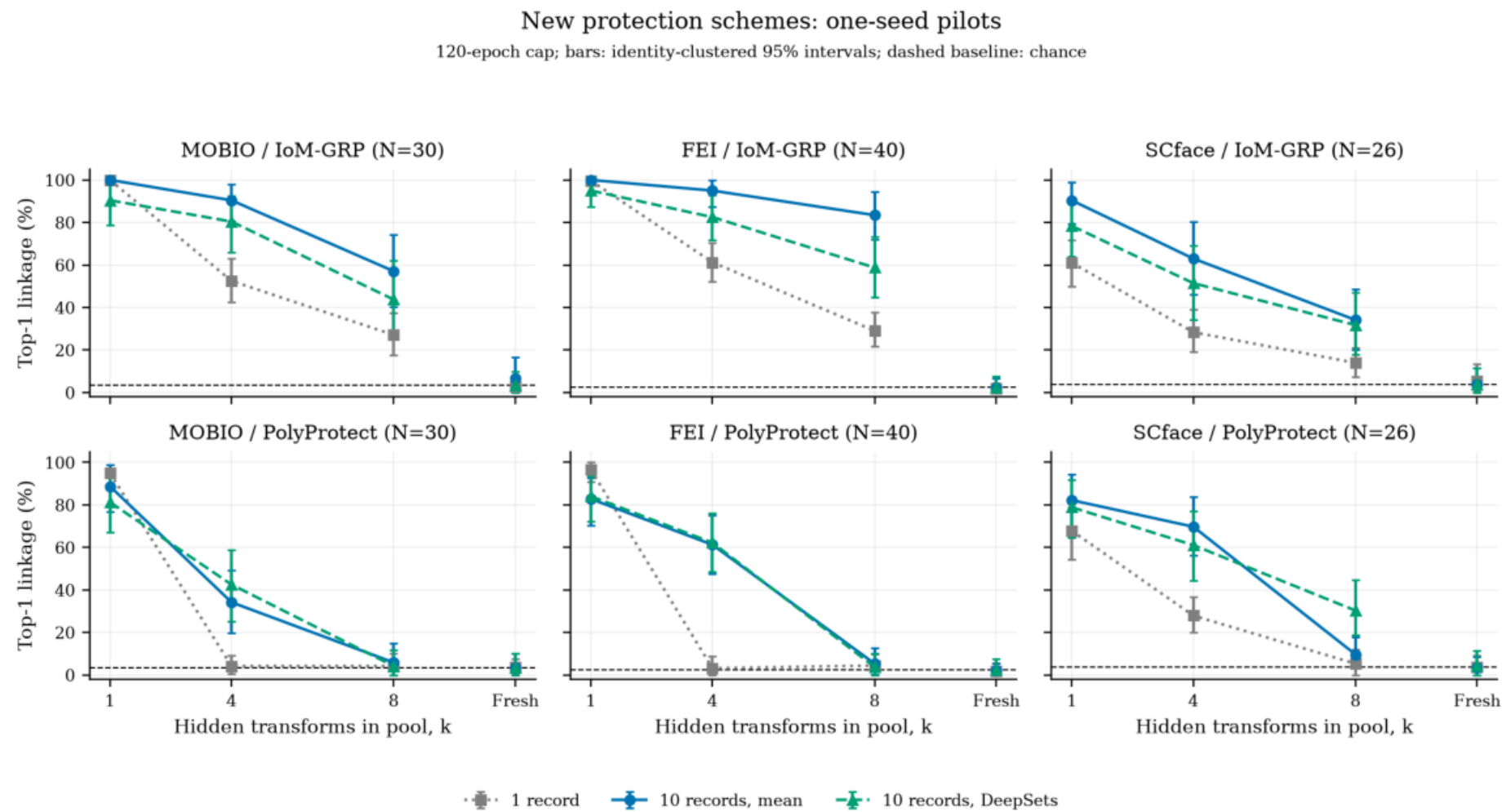
Mechanism and correlation controls

MOBIO mechanism controls (30 gallery identities; chance 3.33%)



Slot identifiers are not key values. Coarse and fine correlation sweeps use separate partitions.

Approved scheme pilots: one seed



Fresh PolyProtect native top-1: MOBIO 12.73%, FEI 13.73%, SCface 10.66%; separate diagnostic.

Ready to draft; not yet ready to submit

Completed: two added datasets, two new schemes, 24 pilot cells, paired intervals.

Before submission:

1. Freeze and authorize staged confirmation; one-seed pilots are not confirmation.
2. Decide whether AgeDB adds enough value for a third added dataset.
3. Approve statistical margins and a multiple-comparison plan.
4. Independently review the corrected theorem and related work.
5. Obtain Sani's scientific and presentation review.

A/A* is a venue ambition, not an established property or an acceptance prediction.